

Coding & Solution

Date	04 July 2026
Team ID	SWTID-2026-2950
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Lakshayjain2006/AI-Powered-Debt-Relief-Financial-Recovery-Platform
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	FastAPI, React.js (Vite), SQLAlchemy
Key Features Implemented	User Registration & Login, JWT Authentication, Loan Management, Financial Health Analysis, Debt-to-Income Ratio, Settlement Prediction, AI Negotiation Letter Generator, Dashboard, AI History
Pending / Incomplete Features	Email Notifications, Credit Score Integration, Payment Gateway, Multi-language Support
Setup / Run Instructions	Install Python 3.11+, Node.js, run <code>pip install -r requirements.txt</code> , <code>npm install</code> , then start FastAPI backend and React frontend using <code>uvicorn main:app --reload</code> and <code>npm run dev</code> .

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The project follows a modular architecture with separate frontend, backend, AI, and database components.
- React.js is used for the responsive user interface, while FastAPI provides secure REST APIs.
- SQLAlchemy ORM manages SQLite database operations efficiently.
- Google Gemini API is integrated for AI-powered financial analysis, settlement prediction, and negotiation letter generation.
- JWT authentication ensures secure user access.
- The codebase is designed to be scalable, maintainable, and reusable for future enhancements.